

Homework 3

September 29, 2024

We have the following code in python for a Tail Recursive QuickSort. Your task is to move the entire code to C and use the -O3 optimization flag to have the fastest possible code:

1. (25 pts) Look how information is passed down the recursion array and (left, right) info. You need to implement this to achieve tail recursion.
2. (25 pts) You need to implement the way of randomly get the uniform probability pivot.
 - (a) Explain why this work
3. (25 pts) Explain the similarities between Python and the Assembly code generated when using -O3
4. (25 pts) Test time complexity between QuickSort using recursion vs QuickSort using TailRecursion.

Note: You can see that

- (a) Exception are used to simulate the use of a single frame for the stack in python.
- (b) The passing of information is necessary to achieve the tail recursion.